

Getein1600 Bi-directional Communication Protocol

1. Getein1600 Data Transmit (Hexadecimal)

Data Frame Format:

Stx Cmd Len Check Block Etx

Stx (0x02)	Frame Head	1 byte
Cmd (0x05)	Command Mode	1 byte
Len	Length	2 bytes
Check	Check	1 byte
Block	Data Block	N bytes
Etx (0x24)	End of Frame	1 byte

Frame Transmit Format:

Stx Cmd Len Check | Barcode | Etx

Example:

Getein1600 transmits frame to upper system: 02 05 0D 00 5B 7C 50 31 32 33 34 35 36 37 38 39 7C 24

Analysis: Check: from beginning to end (0x05 to 0x24); the value is 0x00 before checking and change to 0x5B after checking.

Len: from 0x7C to 0x24, 0x000D bytes in total, displayed in hex (0x0D 0x00).

Block: Barcode (50 31 32 33 34 35 36 3738 39), character representation (P123456789).

2. Server Data Send Back (Character Form)

Data Frame Format:

Stx Cmd Len Block Etx

Stx (#)	Frame Head	1 byte
Cmd (h)	Command Mode	1 byte
Len (00 ~ 99)	Length	2 bytes
Block	Data Block	N bytes
Etx (\$)	End of Frame	1 byte

Data Block Introduction:

Barcode	Barcode, no barcode means blank
ID	Sample ID number
Name	The name of test item
Sample	Sample mode ('0' stands for serum/plasma; '1' stands for whole blood; '2' stands for urine)

Mode	Exhibition Mode ('0' stands for standard mode; '1' stands for quick mode; 0 is system default)
------	--

Frame Transmit Format:

h Len | Barcode | ID | Name | Sample | Mode | \$

Example 1 (Single Item):

Server data frame send back: # h31 | P123456789 | 10 | hs-CRP+CRP | 1 | 0 | \$

Analysis: Len: start from the ' | ', 31 characters in total, displayed as character '3','1'.

Block: Barcode (P123456789), ID (10), Name (hs-CRP+CRP), Sample (1),

Mode (0).

Example 2 (Multiplex Items):

Server data send back:

h 45 | P123456789 | 10 | CK-MB+cTnl+Myo+NT-proBNP | 1 | 0 | \$

Analysis: Len: start from the first ' | ', 45 characters in total, displayed as character '4', '5'.

Block: Barcode (P123456789), ID (10), Name (CK-MB+cTnl, Myo, NT-proBNP),

Sample (1), Mode (0).